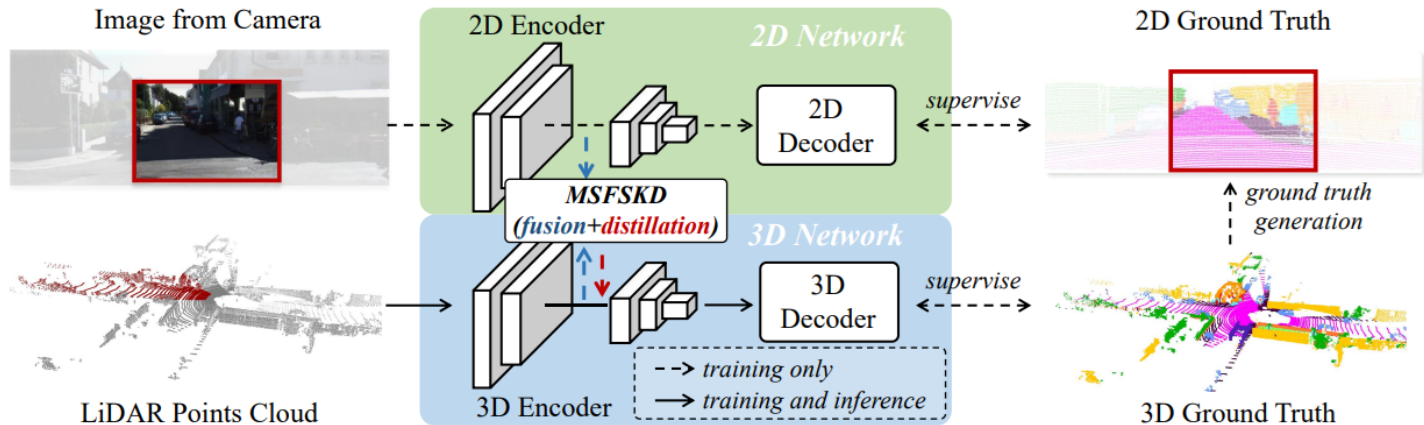


2DPASS: 2D Priors Assisted Semantic Segmentation on LiDAR Point Clouds

ECCV 2022

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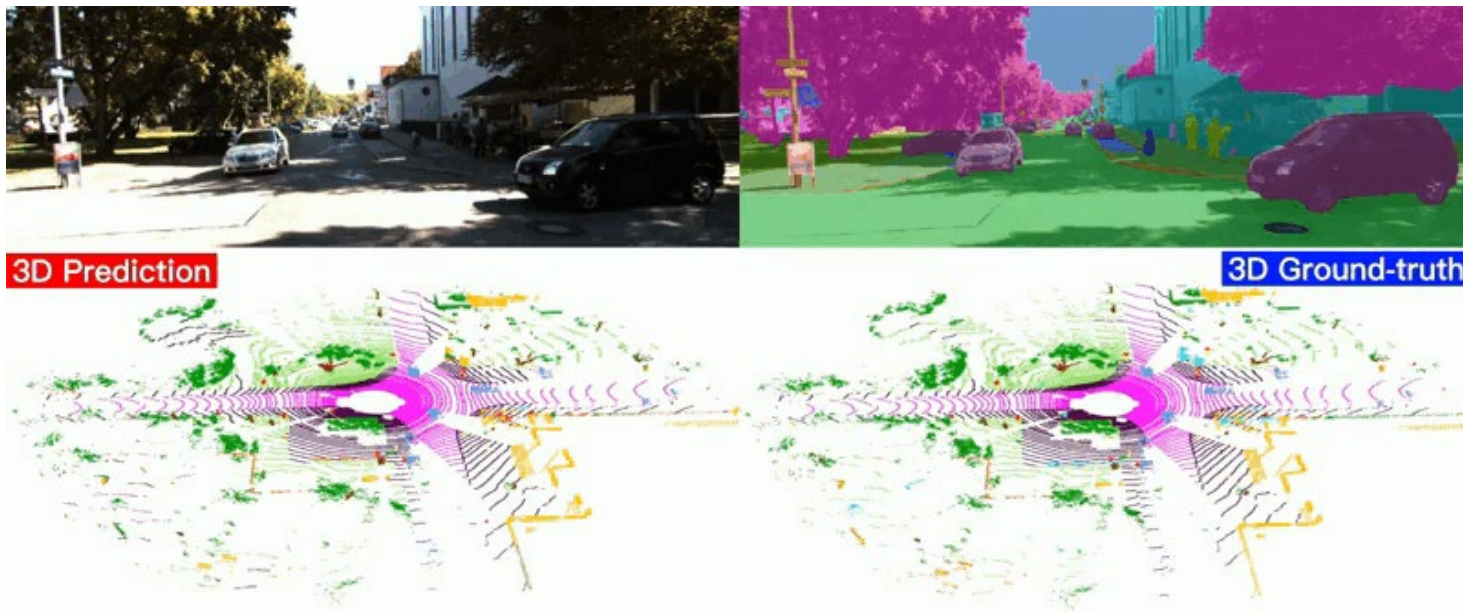


● Problem/Objective

- Semantic Segmentation Enhancement
- Multi-modal Data Fusion and Knowledge Distillation

● Contribution/Key Idea

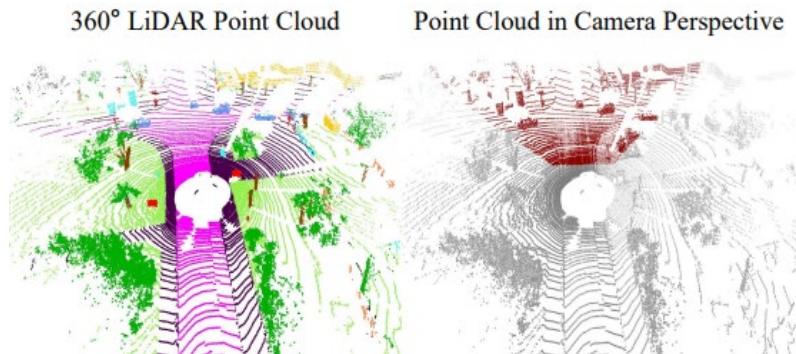
- 2D feature only for training
- First method that distills multi-modal knowledge to single-modal for semantic segmentation



- **Introduction**

- Limitations of Traditional Fusion-Based Approaches

- **FOV difference**
: difference of FOVs between LiDARs & Camera
- **Consume too much Computational resources**
: images + point cloud cost lots



- **Introduction**

- How 2DPASS overcome the Limitations

- **FOV difference**
→ Take images from other time-stamp
- **Consume too much Computational resources**
→ 2D priors images are only used during training

Front-Camera Image and Perspective Projection

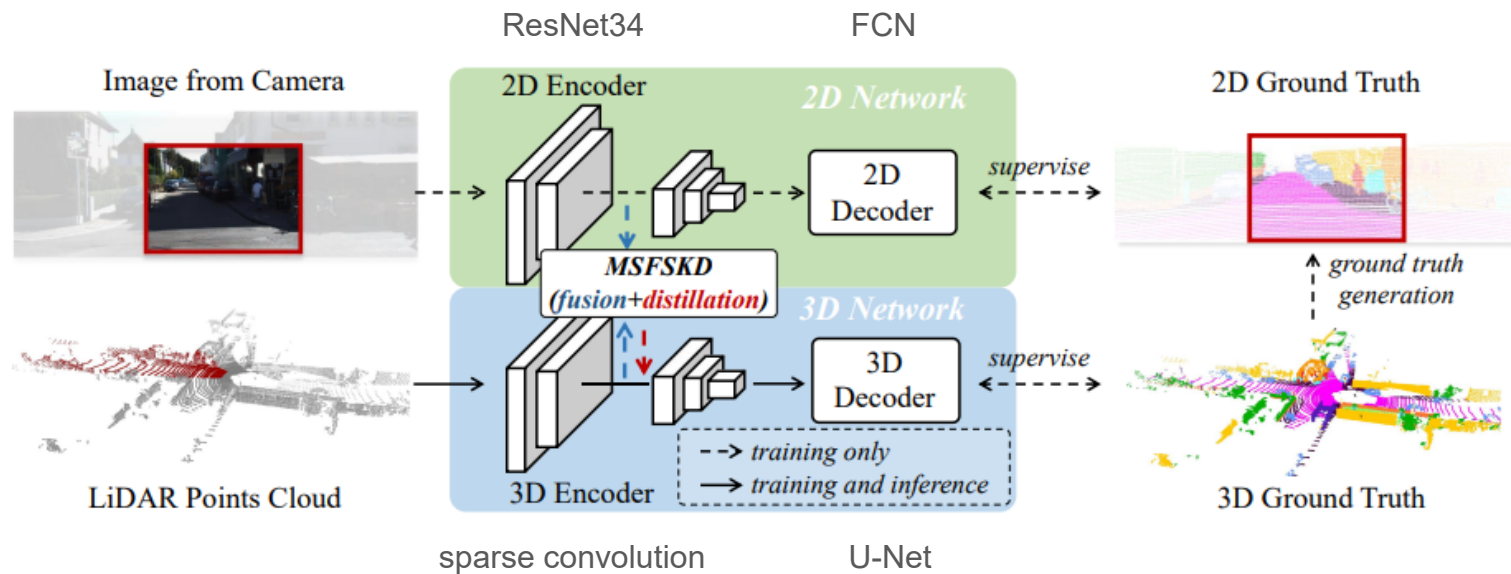


- **Introduction**

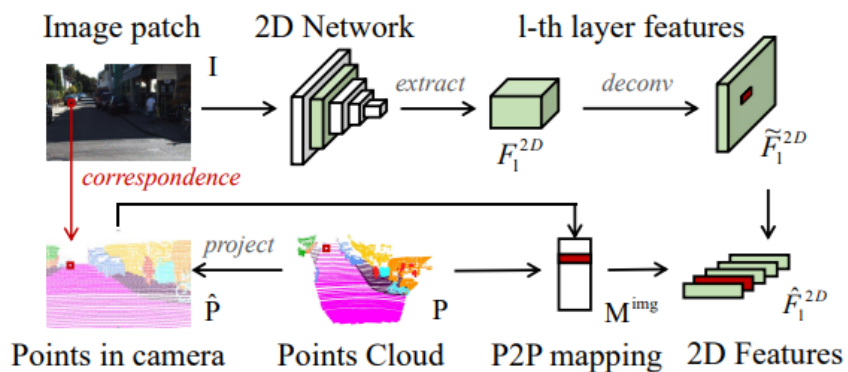
- Main contribution of 2DPASS

- **First method that distills multi-modal knowledge to single point cloud modality for semantic segmentation**
→ 3D LiDAR semantic segmentation with 2D priors from cameras
- **Propose MSFSKD structure**
→ multi-scale fusion-to-single knowledge distillation

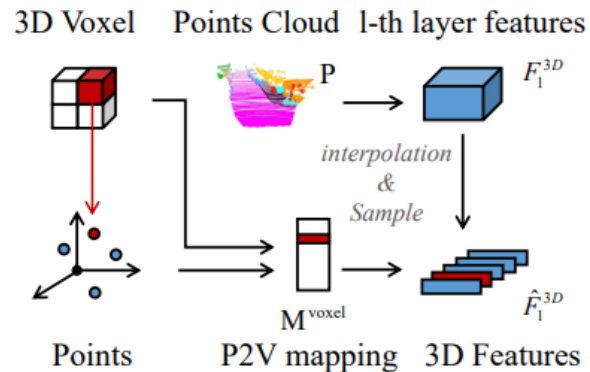
Method - Full 2DPASS structure



Method - 2D & 3D generation



(a) 2D Feature Generation



(b) 3D Feature Generation

$$[u_i, v_i, 1]^T = \frac{1}{z_i} \times K \times T \times [x_i, y_i, z_i, 1]^T$$

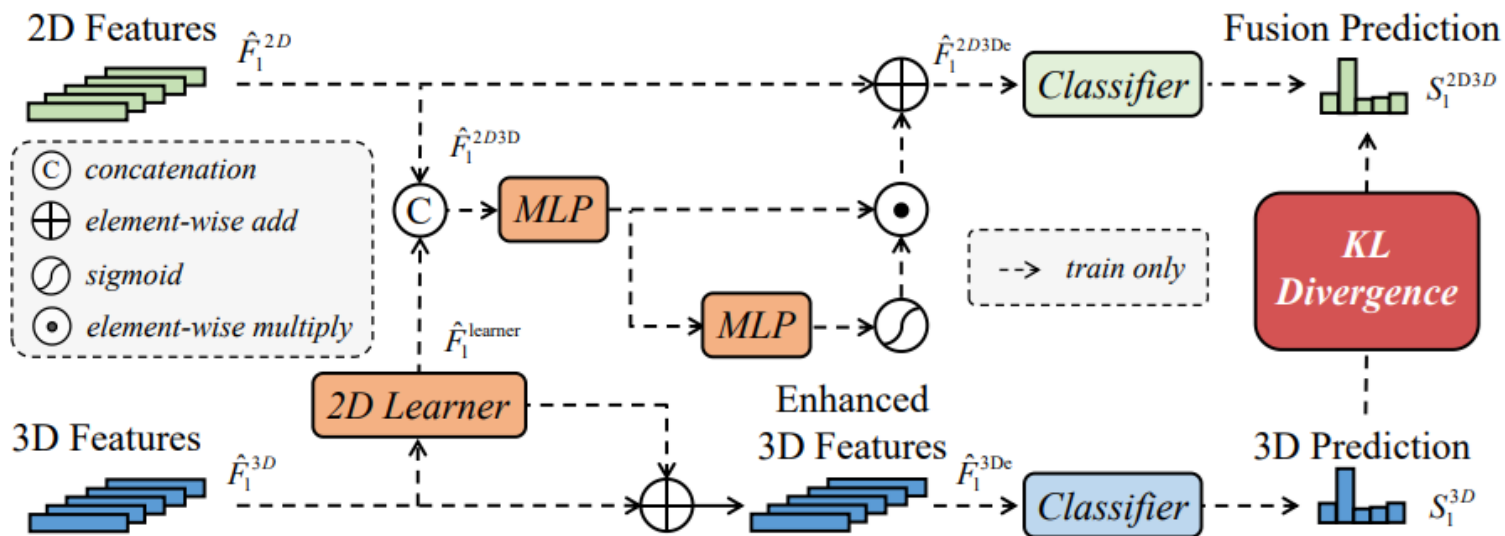
$$M^{img} = \{([v_i], [u_i])\}_{i=1}^N \in \mathbb{R}^{N \times 2}$$

$$\hat{F}^{2D} \in \mathbb{R}^{N^{img} \times D_l}$$

$$M_l^{voxel} = \{([x_i/r_l], [y_i/r_l], [z_i/r_l])\}_{i=1}^N \in \mathbb{R}^{N \times 3}$$

$$\hat{F}_l^{3D} = \{f_i | f_i \in \tilde{F}_l^{3D}, M_{i,1}^{img} \leq H, M_{i,2}^{img} \leq W\}_{i=1}^N \in \mathbb{R}^{N^{img} \times D_l}$$

Method - MSFSKD



$$\hat{F}_l^{2D3De} = \hat{F}_l^{2D} \oplus (\text{MLP}(\hat{F}_l^{2D3D})) \odot \hat{F}_l^{2D3D}$$

$$L_{xM} = D_{KL}(S_l^{2D3D} || S_l^{3D})$$

Experiments

Table 1. Semantic segmentation results on the *SemanticKITTI* test benchmark. Only approaches published before 03/08/2022 are compared.

Method	mIoU	road	sidewalk	parking	other-ground	building	car	truck	bicycle	motorcycle	other-vehicle	vegetation	trunk	terrain	person	bicyclist	motorcyclist	fence	pole	traffic sign	speed (ms)
SqueezeSegV2 [38]	39.7	88.6	67.6	45.8	17.7	73.7	81.8	13.4	18.5	17.9	14.0	71.8	35.8	60.2	20.1	25.1	3.9	41.1	20.2	26.3	-
DarkNet53Seg [16]	49.9	91.8	74.6	64.8	27.9	84.1	86.4	25.5	24.5	32.7	22.6	78.3	50.1	64.0	36.2	33.6	4.7	55.0	38.9	52.2	-
RangeNet53++ [45]	52.2	91.8	75.2	65.0	27.8	87.4	91.4	25.7	25.7	34.4	23.0	80.5	55.1	64.6	38.3	38.8	4.8	58.6	47.9	55.9	83.3
3D-MiniNet [62]	55.8	91.6	74.5	64.2	25.4	89.4	90.5	28.5	42.3	42.1	29.4	82.8	60.8	66.7	47.8	44.1	14.5	60.8	48.0	56.6	-
SqueezeSegV3 [8]	55.9	91.7	74.8	63.4	26.4	89.0	92.5	29.6	38.7	36.5	33.0	82.0	58.7	65.4	45.6	46.2	20.1	59.4	49.6	58.9	238
PointNet++ [25]	20.1	72.0	41.8	18.7	5.6	62.3	53.7	0.9	1.9	0.2	0.2	46.5	13.8	30.0	0.9	1.0	0.0	16.9	6.0	8.9	5900
TangentConv [36]	40.9	83.9	63.9	33.4	15.4	83.4	90.8	15.2	2.7	16.5	12.1	79.5	49.3	58.1	23.0	28.4	8.1	49.0	35.8	28.5	3000
PointASNL [31]	46.8	87.4	74.3	24.3	1.8	83.1	87.9	39.0	0.0	25.1	29.2	84.1	52.2	70.6	34.2	57.6	0.0	43.9	57.8	36.9	-
RandLA-Net [1]	55.9	90.5	74.0	61.8	24.5	89.7	94.2	43.9	29.8	32.2	39.1	83.8	63.6	68.6	48.4	47.4	9.4	60.4	51.0	50.7	880
KPConv [29]	58.8	90.3	72.7	61.3	31.5	90.5	95.0	33.4	30.2	42.5	44.3	84.8	69.2	69.1	61.5	61.6	11.8	64.2	56.4	47.4	-
PolarNet [63]	54.3	90.8	74.4	61.7	21.7	90.0	93.8	22.9	40.3	30.1	28.5	84.0	65.5	67.8	43.2	40.2	5.6	61.3	51.8	57.5	62
JS3C-Net [2]	66.0	88.9	72.1	61.9	31.9	92.5	95.8	54.3	59.3	52.9	46.0	84.5	69.8	67.9	69.5	65.4	39.9	70.8	60.7	68.7	471
SPVNAS [10]	67.0	90.2	75.4	67.6	21.8	91.6	97.2	56.6	50.6	50.4	58.0	86.1	73.4	71.0	67.4	67.1	50.3	66.9	64.3	67.3	259
Cylinder3D [40]	68.9	92.2	77.0	65.0	32.3	90.7	97.1	50.8	67.6	63.8	58.5	85.6	72.5	69.8	73.7	69.2	48.0	66.5	62.4	66.2	131
RPVNet [42]	70.3	93.4	80.7	70.3	33.3	93.5	97.6	44.2	68.4	68.7	61.1	86.5	75.1	71.7	75.9	74.4	43.4	72.1	64.8	61.4	168
(AF) ² -S3Net [41]	70.8	92.0	76.2	66.8	45.8	92.5	94.3	40.2	63.0	81.4	40.0	78.6	68.0	63.1	76.4	81.7	77.7	69.6	64.0	73.3	-
Baseline	67.4	89.8	73.8	62.1	33.5	91.9	96.3	54.9	51.1	55.8	51.6	86.5	72.3	71.3	76.8	79.8	30.3	68.7	63.7	70.2	62
2DPASS(Ours)	72.9	89.7	74.7	67.4	40.0	93.5	97.0	61.1	63.6	63.4	61.5	86.2	73.9	71.0	77.9	81.3	74.1	72.9	65.0	70.4	62

Experiments

Table 2. Comparison to the state-of-the-art methods on the test set of SemanticKITTI multiple scans challenge. -s indicates static and -m stands for moving.

Method	mIoU	Acc	car-s	car-m	truck-s	truck-m	other-s	other-m	person-s	person-m	bicyclist-s	bicyclist-m	motorcyclist-s	motorcyclist-m
LatticeNet [64]	45.2	89.3	91.1	54.8	29.7	3.5	23.1	0.6	6.8	49.9	0.0	44.6	0.0	64.3
TemporalLidarSeg [65]	47.0	89.6	92.1	68.2	39.2	2.1	35.0	12.4	14.4	40.4	0.0	42.8	0.0	12.9
KPConv [29]	51.2	89.3	93.7	69.4	42.5	5.8	38.6	4.7	21.6	67.5	0.0	67.4	0.0	47.2
Cylinder3D [40]	52.5	91.0	94.6	74.9	41.3	0.0	38.8	0.1	12.5	65.7	1.7	68.3	0.2	11.9
(AF) ² -S3Net [41]	56.9	88.1	91.8	65.3	15.7	5.6	27.5	3.9	16.4	67.6	15.1	66.4	67.1	59.6
2DPASS(Ours)	62.4	91.4	96.2	82.1	48.2	16.1	52.7	3.8	35.4	80.3	7.9	71.2	62.0	73.1

- Experiments

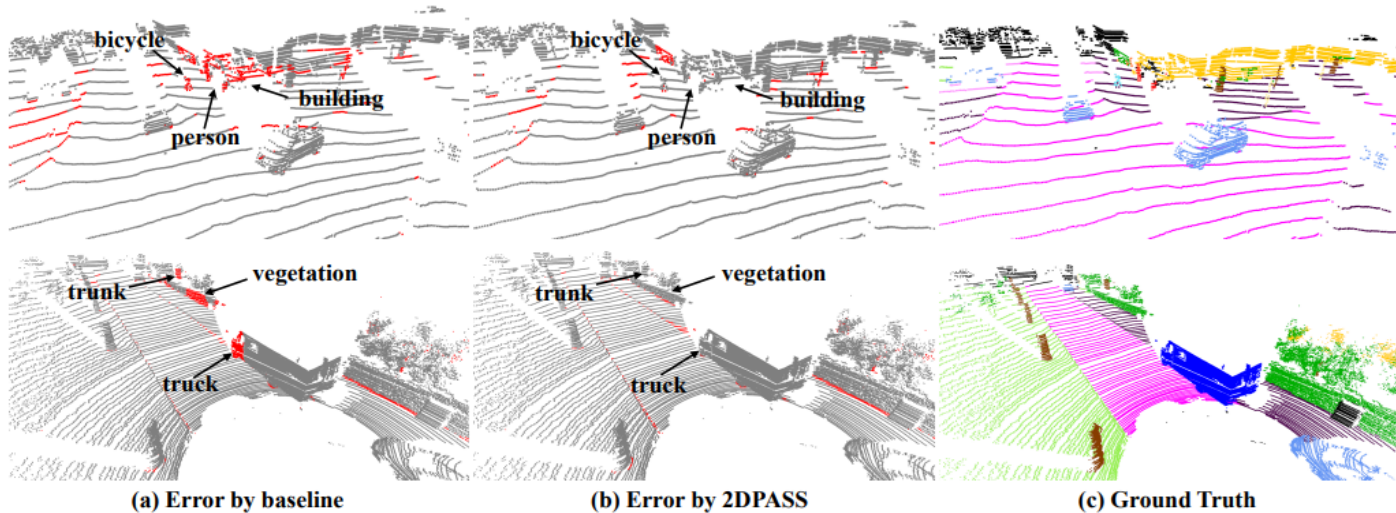


Fig. 5. Qualitative results of 2DPASS on the validation set of SemanticKITTI. Our baseline has a higher error recognizing small objects and region boundaries, while 2DPASS recognizes small objects better thanks to the prior of 2D modality.

Experiments

Table 3. Semantic segmentation results on the *Nuscenes* test benchmark. Only approaches published before 03/08/2022 are compared. *L* and *C* stand for LiDAR and camera, respectively. (*) The speed reported in PMF [13] is accelerated by TensorRT, and we test their model without such technique in the same environment.

Method	Input	mIoU	FW mIoU	barrier	bicycle	bus	car	construction	motorcycle	pedestrian	traffic cone	trailer	truck	driveable	other flat	sidewalk	terrain	manmade	vegetation	speed (ms)
PolarNet [63]	L	69.4	87.4	72.2	16.8	77.0	86.5	51.1	69.7	64.8	54.1	69.7	63.5	96.6	67.1	77.7	72.1	87.1	84.5	-
JS3C-Net [2]	L	73.6	88.1	80.1	26.2	87.8	84.5	55.2	72.6	71.3	66.3	76.8	71.2	96.8	64.5	76.9	74.1	87.5	86.1	-
Cylinder3D [40]	L	77.2	89.9	82.8	29.8	84.3	89.4	63.0	79.3	77.2	73.4	84.6	69.1	97.7	70.2	80.3	75.5	90.4	87.6	63
AMVNet [39]	L	77.3	90.1	80.6	32.0	81.7	88.9	67.1	84.3	76.1	73.5	84.9	67.3	97.5	67.4	79.4	75.5	91.5	88.7	85
SPVCNN [10]	L	77.4	89.7	80.0	30.0	91.9	90.8	64.7	79.0	75.6	70.9	81.0	74.6	97.4	69.2	80.0	76.1	89.3	87.1	63
(AF) ² -S3Net [41]	L	78.3	88.5	78.9	52.2	89.9	84.2	77.4	74.3	77.3	72.0	83.9	73.8	97.1	66.5	77.5	74.0	87.7	86.8	270
PMF [13]	L+C	77.0	89.0	82.0	40.0	81.0	88.0	64.0	79.0	80.0	76.0	81.0	67.0	97.0	68.0	78.0	74.0	90.0	88.0	125*
2D3DNet [66]	L+C	80.0	90.1	83.0	59.4	88.0	85.1	63.7	84.4	82.0	76.0	84.8	71.9	96.9	67.4	79.8	76.0	92.1	89.2	-
Baseline	L	77.6	88.5	80.8	37.9	92.7	90.5	65.4	77.6	71.5	70.9	83.1	75.3	97.0	69.3	78.1	75.6	89.1	86.8	44
2DPASS(Ours)	L	80.8	90.1	81.7	55.3	92.0	91.8	73.3	86.5	78.5	72.5	84.7	75.5	97.6	69.1	79.9	75.5	90.2	88.0	44

● Experiments

Table 4. Comparison with different knowledge distillation.

Method	SemanticKITTI
Hinton <i>et.al.</i> [46]	66.34
Huang <i>et.al.</i> [67]	66.46
Yang <i>et.al.</i> [68]	66.75
xMUDA [17]	67.88
2DPASS	69.32

Table 5. Ablation study on the SemanticKITTI validation set.

baseline	MSFSKD				SemanticKITTI
	KL Div	Modality Fusion	2D Learner		
✓					65.58
✓	✓				66.34
✓	✓	✓			69.13
✓	✓	✓	✓		69.32